UNIT 4 MINI PROJECT REPORT

Introduction to Networking and Data Communication

Project Title:

Configure DHCP and Set Up a Network Using Cisco Packet Tracer

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Class: SY Cyber Security

Subject: Network Protocol Configuration

1. Aim

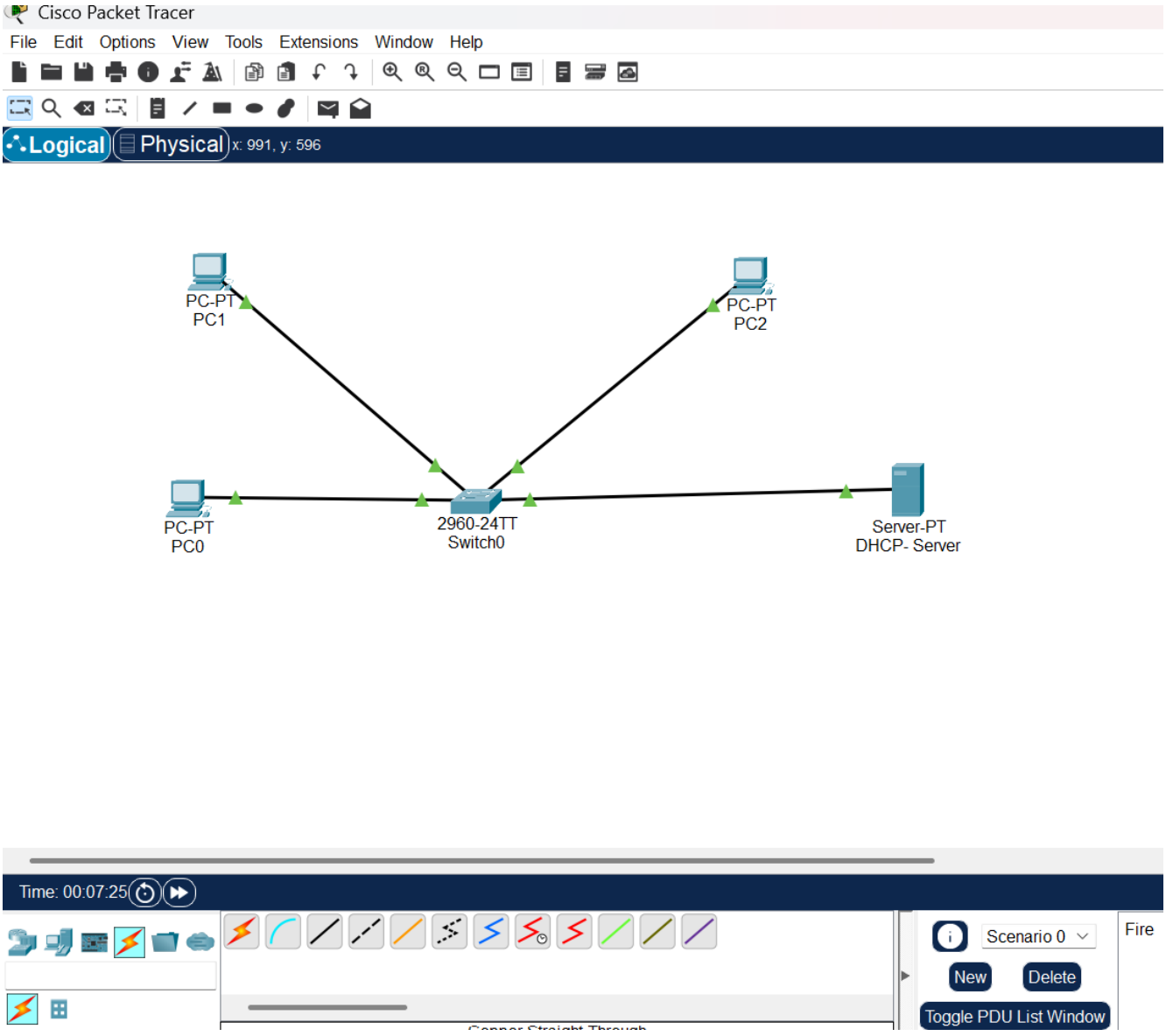
To configure a DHCP server in Cisco Packet Tracer and test automatic IP configuration and connectivity between PCs on a local network.

2. Requirements

- Cisco Packet Tracer
- One 2960 switch
- Three PCs (PC0, PC1 and PC2)
- One Server-PT device
- Copper Straight-Through cables

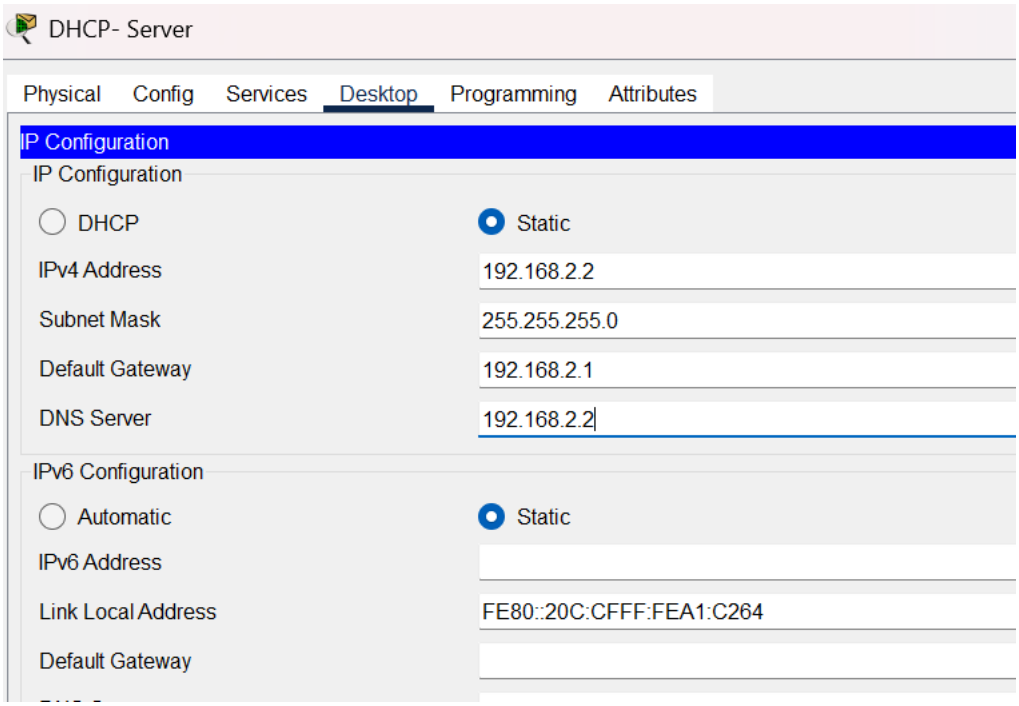
3. Network Topology

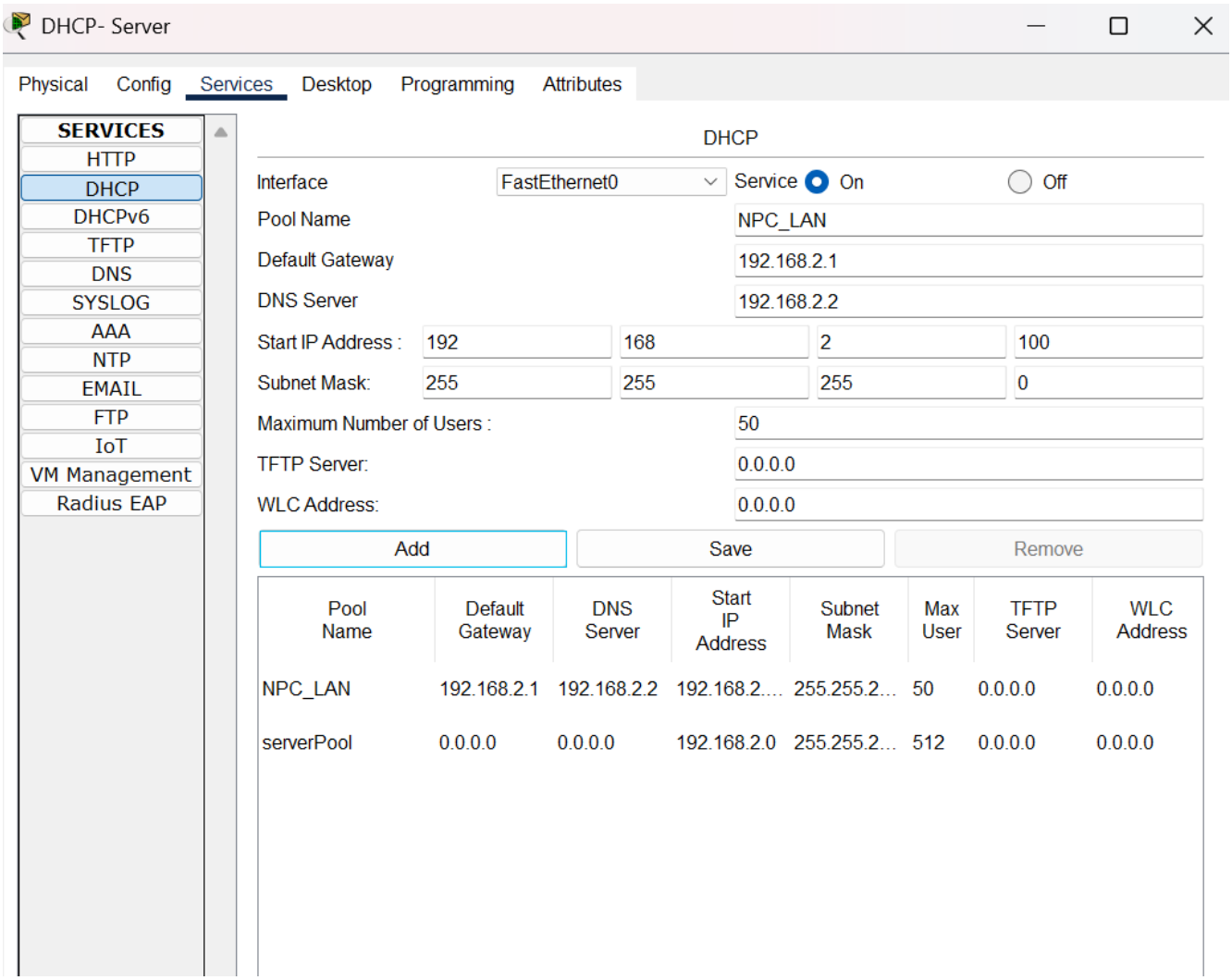
PC0, PC1, PC2 and the DHCP server are connected to Switch0 using Copper Straight-Through cables.



4. IP Addressing and DHCP Configuration

Device/Setting	Configuration
DHCP Server IP	192.168.2.2
Subnet Mask	255.255.255.0
Pool Name	NPC_LAN
Default Gateway	192.168.2.1
DNS Server	192.168.2.2
Starting IP Address	192.168.2.100
Maximum Users	50


DHCP- Server
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
192.168.2.2
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20C:CFFF:FEA1:C264
Default Gateway


DHCP- Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
NPC_LAN
Default Gateway
192.168.2.1
DNS Server
192.168.2.2
Start IP Address :
192
168
2
100
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
NPC_LAN
192.168.2.1
192.168.2.2
192.168.2....
255.255.2...
50
0.0.0.0
0.0.0.0
serverPool
0.0.0.0
0.0.0.0
192.168.2.0
255.255.2...
512
0.0.0.0
0.0.0.0

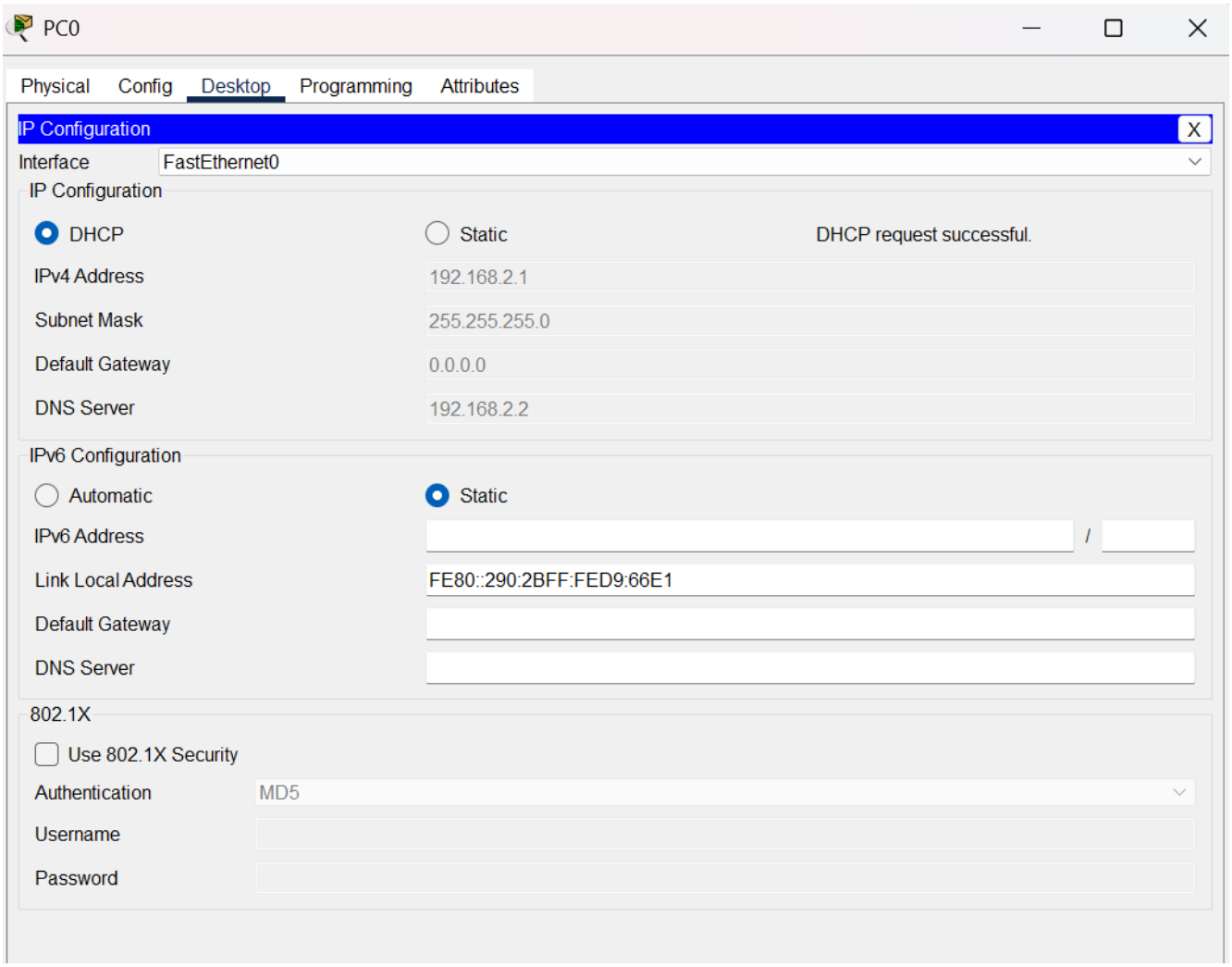
5. Procedure

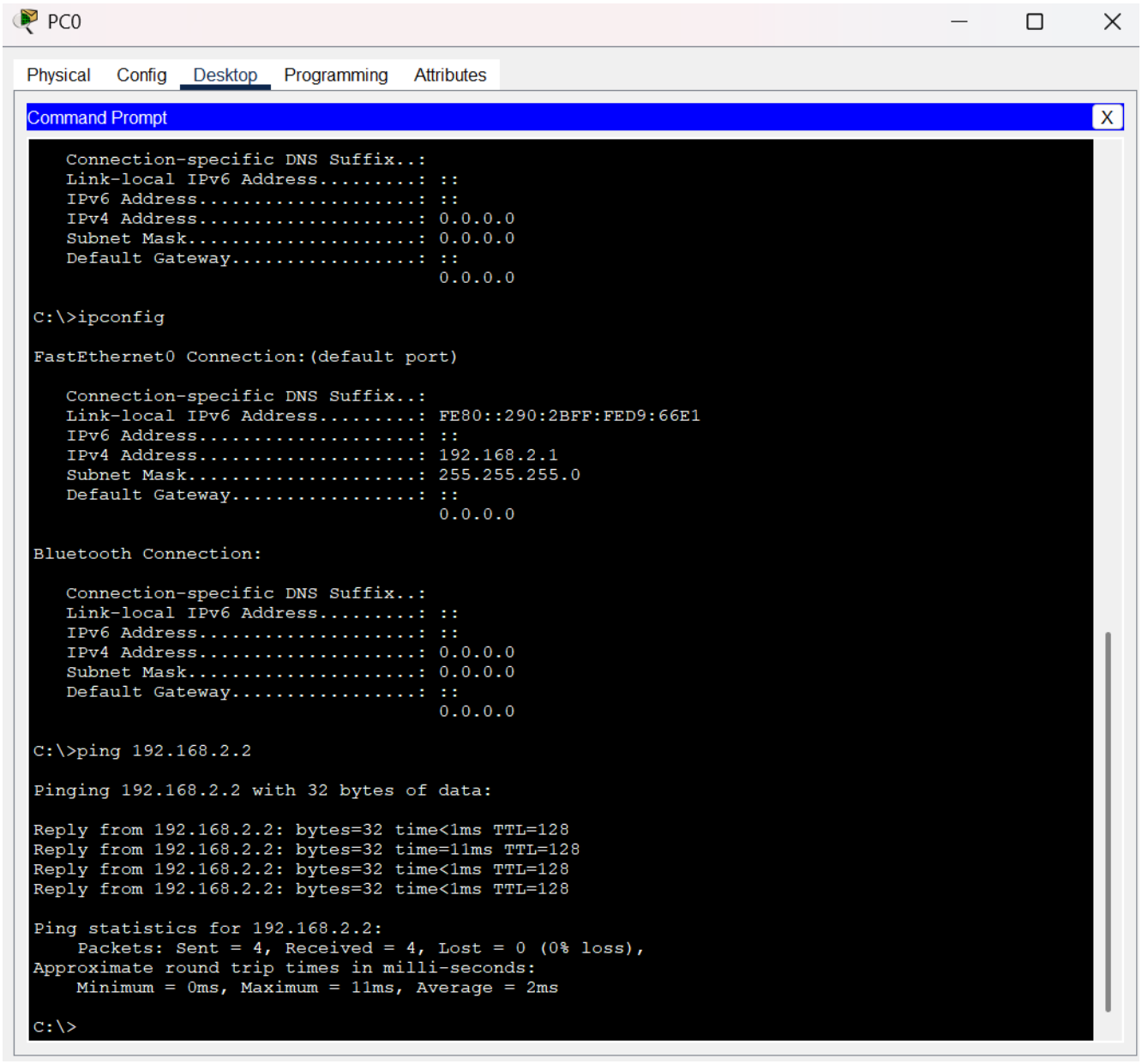
1. Open Cisco Packet Tracer and place three PCs, one server and one 2960 switch.
2. Connect all devices to the switch using Copper Straight-Through cables.
3. Open the server's **Desktop** → **IP Configuration** and configure the static IP address as 192.168.2.2 with subnet mask 255.255.255.0.
4. Open **Server** → **Services** → **DHCP** and turn the DHCP service On.
5. Create the DHCP pool named NPC_LAN. Enter the gateway, DNS server, starting IP address, subnet mask and maximum number of users listed above. Click **Add**.
6. Open each PC's **Desktop** → **IP Configuration** and select **DHCP**. Do not enter static IP addresses on the PCs.
7. Open each PC's Command Prompt and run ipconfig to check the assigned IP address.
8. Test connectivity using the ping command and the actual IP address of the destination device.

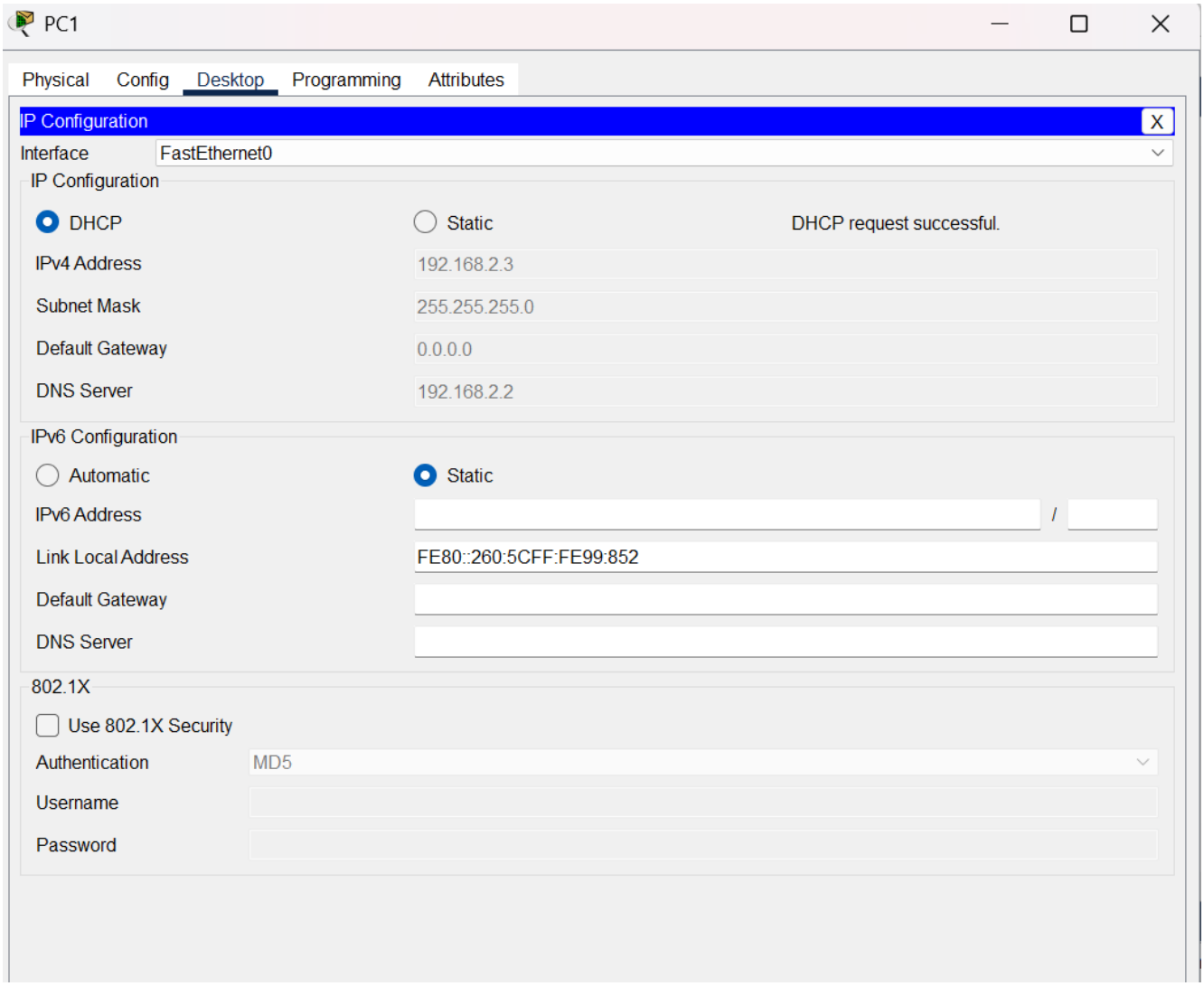
6. Connectivity Testing

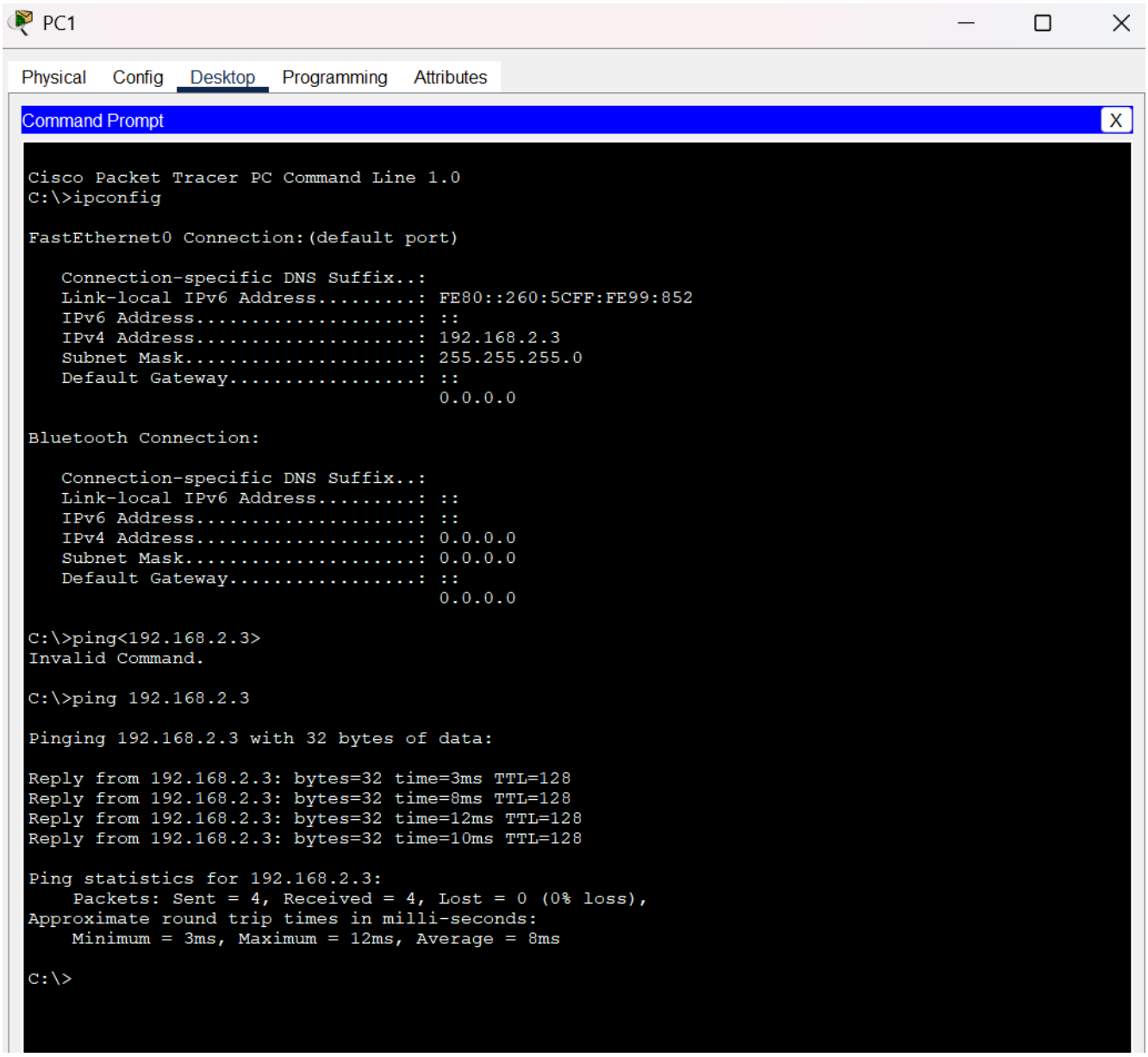
The following ping tests were completed successfully:

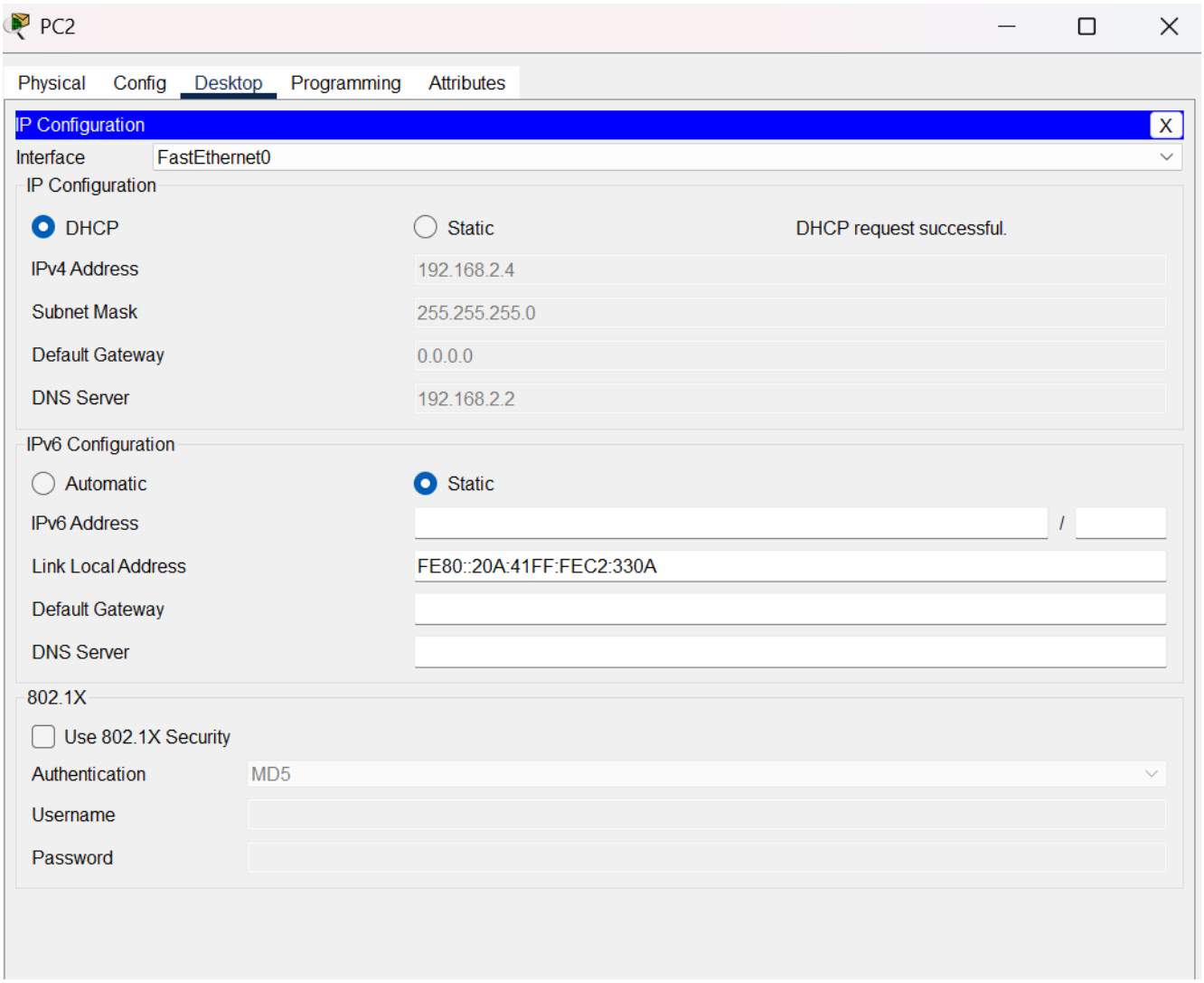
- PC0 to DHCP server: ping 192.168.2.2
- PC0 to PC1: ping 192.168.2.3
- PC0 to PC2: Use the actual IP address shown by PC2.

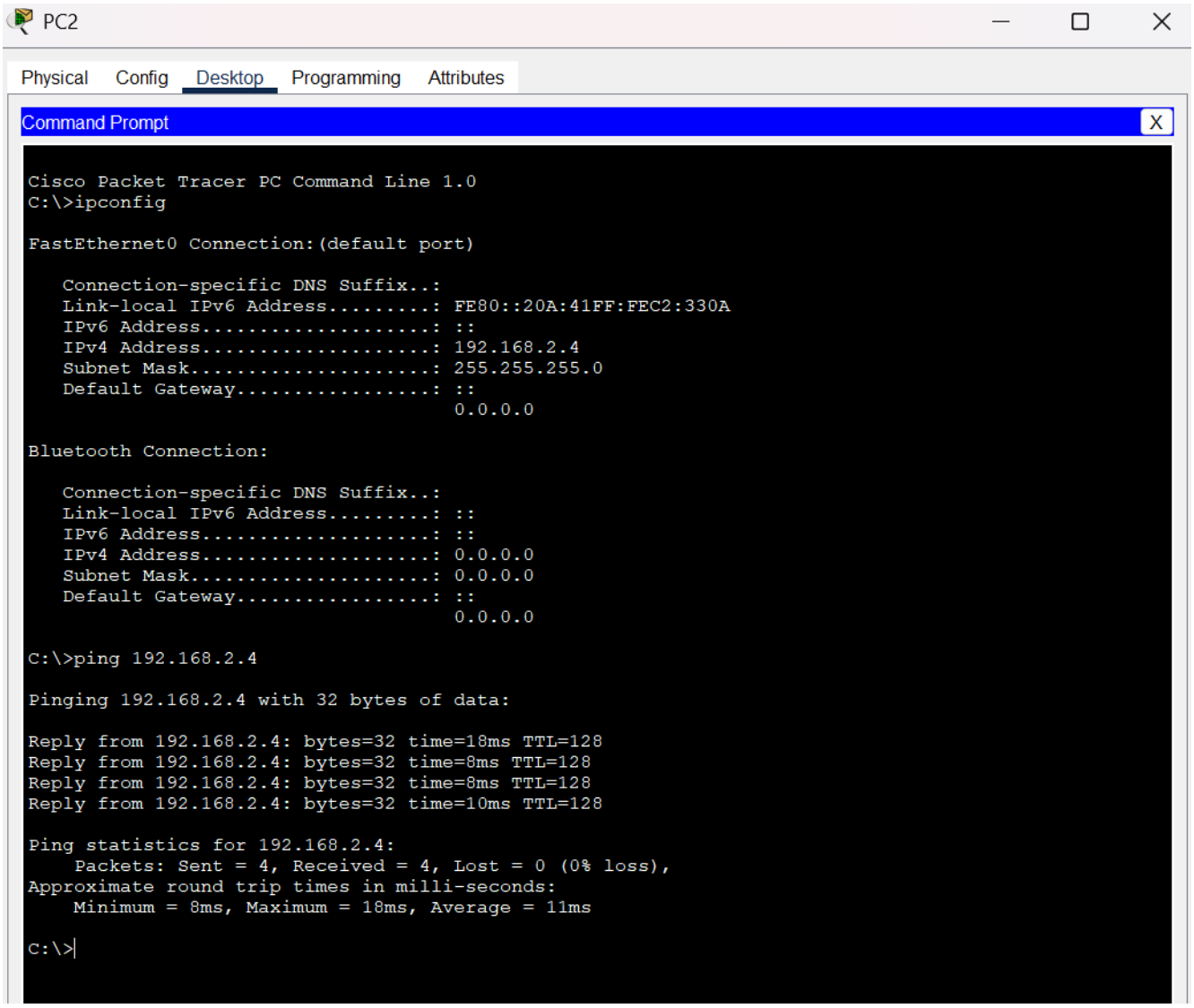

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.2.1
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
192.168.2.2
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::290:2BFF:FED9:66E1
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::290:2BFF:FED9:66E1
IPv6 Address.....: ::
IPv4 Address.....: 192.168.2.1
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=11ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 2ms
C:\>


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.2.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
192.168.2.2
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:5CFF:FE99:852
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::260:5CFF:FE99:852
IPv6 Address.....: ::
IPv4 Address.....: 192.168.2.3
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping<192.168.2.3>
Invalid Command.
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128
Reply from 192.168.2.3: bytes=32 time=8ms TTL=128
Reply from 192.168.2.3: bytes=32 time=12ms TTL=128
Reply from 192.168.2.3: bytes=32 time=10ms TTL=128
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 12ms, Average = 8ms
C:\>


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.2.4
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
192.168.2.2
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20A:41FF:FEC2:330A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::20A:41FF:FEC2:330A
IPv6 Address.....: ::
IPv4 Address.....: 192.168.2.4
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.2.4
Pinging 192.168.2.4 with 32 bytes of data:
Reply from 192.168.2.4: bytes=32 time=18ms TTL=128
Reply from 192.168.2.4: bytes=32 time=8ms TTL=128
Reply from 192.168.2.4: bytes=32 time=8ms TTL=128
Reply from 192.168.2.4: bytes=32 time=10ms TTL=128
Ping statistics for 192.168.2.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 18ms, Average = 11ms
C:\>

Observation: PC0 displayed 192.168.2.1 and PC1 displayed 192.168.2.3 during testing. These addresses do not match the configured pool's starting address of 192.168.2.100. Therefore, successful ping tests confirm connectivity, but DHCP pool allocation still needs verification. The PCs should remain configured for DHCP.

7. Result

The network topology was created in Cisco Packet Tracer, the DHCP service and NPC_LAN pool were configured, and successful ping replies were observed between PC0, the server and the other PCs. The displayed IP addresses should be checked against the configured DHCP pool before claiming that automatic IP assignment has been fully verified.

8. Conclusion

DHCP simplifies network administration by automatically providing IP configuration to client devices. Ping testing confirms local connectivity between devices on the same subnet. A router is required for communication outside this local network.

— End of Report —